

Five-Stage Fourth-Order SDIRK

Skvortsov S54b non-geometric reference method

1 Butcher tableau

For $\dot{z} = f(t, z)$, the S54b tableau is

$$\begin{array}{c|ccccc} \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{8} & \frac{3}{8} & \frac{1}{4} & 0 & 0 \\ 1 & -\frac{3}{2} & \frac{3}{4} & \frac{2}{3} & \frac{1}{4} & 0 \\ 1 & 0 & \frac{1}{6} & \frac{2}{3} & -\frac{1}{12} & \frac{1}{4} \\ \hline & 0 & \frac{1}{6} & \frac{2}{3} & -\frac{1}{12} & \frac{1}{4} \end{array}.$$

All five diagonal entries equal $\gamma = 1/4$, so the method is singly diagonally implicit. The final row of A equals b^T , making the method stiffly accurate.

The stages are solved sequentially:

$$Z_i = z_n + h \sum_{j < i} a_{ij} f(t_n + c_j h, Z_j) + h \gamma f(t_n + c_i h, Z_i), \quad (1)$$

$$z_{n+1} = z_n + h \sum_{i=1}^5 b_i f(t_n + c_i h, Z_i). \quad (2)$$

Each stage therefore requires a nonlinear solve in the physical state dimension, rather than one simultaneous system containing all stages.

2 Fourth-order conditions

Writing $C = \text{diag}(c)$ and using componentwise powers, the coefficients satisfy

$$\begin{aligned} b^T \mathbf{1} &= 1, & b^T c &= \frac{1}{2}, & b^T c^2 &= \frac{1}{3}, & b^T A c &= \frac{1}{6}, \\ b^T c^3 &= \frac{1}{4}, & b^T C A c &= \frac{1}{8}, & b^T A c^2 &= \frac{1}{12}, & b^T A^2 c &= \frac{1}{24}. \end{aligned}$$

These are the Runge–Kutta conditions through order four. For sufficiently smooth problems the local truncation error is $\mathcal{O}(h^5)$ and the global error is $\mathcal{O}(h^4)$.

3 Deliberately non-geometric structure

A Runge–Kutta method is symplectic for general canonical Hamiltonian systems only if

$$b_i a_{ij} + b_j a_{ji} - b_i b_j = 0 \quad \text{for every } i, j.$$

The S54b coefficients do not satisfy this identity. For example, with $i = j = 5$ the left-hand side is $1/16$, not zero. Hence S54b is not a symplectic Runge–Kutta method.

Symmetry would require equality with the adjoint tableau, including

$$c_i = 1 - c_{6-i}, \quad b_i = b_{6-i}, \quad a_{ij} + a_{6-i, 6-j} = b_{6-j}.$$

Already $c_1 = 1/4 \neq 1 - c_5 = 0$, so S54b is not symmetric. These failures make the method a useful same-order control in long-time comparisons against geometric fourth-order methods.

4 Newton implementation

For a known right-hand side r_i , one stage solves

$$R_i(Z_i) = Z_i - r_i - h\gamma f(t_n + c_i h, Z_i) = 0.$$

Newton corrections satisfy

$$\left[I - h\gamma D_z f(t_n + c_i h, Z_i^{(k)}) \right] \delta_i^{(k)} = -R_i(Z_i^{(k)}).$$

For guiding-centre dynamics the implementation uses independent analytic 2×2 particle Jacobians. A centered finite-difference dense Jacobian is available for generic dynamical systems. Per-stage and per-step iterations, residual evaluations, residual norms, and effective tolerances are returned as diagnostics.

5 Reference

The coefficient table is the S54b method presented by L. M. Skvortsov, “Diagonally implicit Runge–Kutta methods for stiff problems,” *Computational Mathematics and Mathematical Physics*, 46(12), 2110–2123 (2006), DOI: 10.1134/S0965542506120098.